

Gesture Controlled Bomb Diffusing Mobile Robot

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Abstract — Hand gesture control system has been unfolding the entire humanity to quite a wide range of applications, such as human android interaction, automation & control, virtual games, 3D animations, posture recognition language, traffic signal controlling, virtual play station, etc. This glove can be used to control small scale machines and home appliances. Increased usage of robots in unfavorable cases where conditions don't stay certain like rescue operations and nuclear environment has made advancement in the modern world of army and defense. This work proposes an integrated approach of tracking and recognition of hands which is intended to be used as human-robot interaction interface (inferred from HMI (human machine interface)).

Keywords - Arduino UNO, Arduino DUE, Accelerometer, Flex Sensors, Electromagnetic Pulse Gun(EMP), Human Machine Interface(HMI)

I. INTRODUCTION

Gesture is seen as a sequence of successive postures. Postures are a set of feasible movements of the fingers, the data from the hand and an additional value which indicates the re-habitability of the posture and its value. Gesture recognition technologies are much younger in today's world[10]. Introduction of sensors like accelerometer and flex sensor has taken the field of robotics into a great run. In this system, encoded data is given to the microcontroller through the transmitter, which later is decoded by the receiver and the posture is recognized. The introduced system includes an accelerometer-based technique which controls a far placed factory robot via. two three-axis wired accelerometers. The accelerometers are fixed to the arm of a user capturing the system's gesture. Movement of the robot and arm can also be monitored using a Wi-Fi camera which is mounted on the robotic arm along with its EMP and gripper function. In addition, to a great demand in this field, the system is expected to be controlled by the militants (especially) in a most efficient and cost-effective way.

II. LITERATURE SURVEY

Robotics is ruling almost all the domains like health, military, home automation, etc. Introduction of robots influenced industrial field drastically. Continuous development of robotics attracted most of the engineers. One such purpose of

introduction of robot is to perform the repetitive task which cannot be done by the humans because of physical and mental limitations[1]. Not only this task, robots can also be used to perform certain tasks where humans risk to enter[12]. Like, on sep 12 ,2001 in United States of America , there was a major loss of life because of terrorist attack on World Trade Centre. Robots were used for rescue operations like locating the people present in the World Trade Centre by controlling from distant. Those robots had network cables that were trailing behind them, which were used to transmit images from robot. Wired robots has its own limitations[2]. Longer wires should be used for the robots for controlling, this increases the cost of robot. By the introduction of wireless communication like Bluetooth, Zigbee etc, wireless control of robots had become easier but was restricted to smaller distances. Range of operation can be increased by changing the communication medium interfaced between user (Joystick and Gesture User Interface(GUI)) and robot. This introduction of robots replaced so many people in the security forces by doing the activities like monitoring, rescuing, attacking the enemy forces, bomb detection and deactivation etc. This reduced the probability of loss of human life [3]. With a variety of usage on robotics field, a pure human to android interaction is necessary in order to safeguard our mankind from any kind of hazardous environment. Appropriate gestures are given to the system which helps interface to the user at variable distant too [4]. Introduction of sensors like accelerometer and flex sensor has improvised the field of robotics to a great turn. This sensors replaced joysticks which has its own limitations like heavy weight, mechanical losses etc.[5]. The paper considered represents development of a multipurpose wireless jacket which helps control a robotic arm by accepting the human actions[8]. The final system is consists of a wearable jacket that posses flex sensors which helps track the user's behavior [6].

III. THEORETICAL INFRASTRUCTURE:

The model below consists of 2 parts i.e. transmitter end and receiver end. The transmitter end consists of a pair of gloves (embedded with flex sensors and accelerometer) and Arduino UNO which intakes the encoded gestures from the sensors and accelerometers. The receiver end includes a robotic arm mounted on a robot which is directed by the decoded gestures from the user. In addition, it also poses an EMP gun (acts as a jammer in case of any default while diffusion) and a wireless camera (which directs the user in controlling the robot).

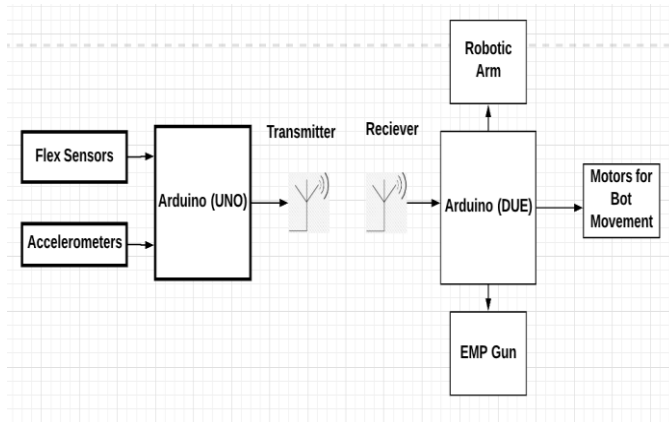


Fig 1: Typical block diagram

The objective of the entire system can be detailed as follows,

i. To build efficient human-machine interaction
 Gesture-based input devices apply a manual interaction between the human and the machine[7]. They provide a base in understanding this field by applying key hand gesture tracking techniques and applications using hand glove-based input.

ii. To build a human rescue robot

Robots work under unfavorable conditions, for search and rescue after disasters like nuclear environments, fire hazards, etc. Robots are basically been preferred to use in environments that are too hazardous for human life to enter. Detailed information of implementation and working of the prototype is discussed along with the individual components description. Further, the interconnection between the components and the microcontroller is also illustrated legitimately below. Categorizing the system into two parts;

1. Transmitter end:

a. Flex sensors:

Flex sensor is a varying resistive device used to detect bents or flexes. The output data from the sensor is analog which is sent to the Arduino UNO from where it is further forwarded to the receiver end.

b. Accelerometer (ADXL345):

Accelerometer works on the principle of Micro Electrical Mechanical Systems (MEMS). It is a thin layered functional device which records the three dimensional data from the surrounding deflection[12]. It has a higher resolution (13 bit). It is mostly used in handsets, medical instruments, gaming devices, industrial instruments, fitness equipment[11]. The acquired values from two accelerometers on the either hands will be given to the Arduino UNO.

c. Arduino UNO:

Arduino UNO is an ATmega328P based microcontroller. It consists of 16MHz quartz crystal. Its operating voltage is 5V consisting of flash memory of 32kB and SRAM of 2kB. This controller is used at the transmitter side from where the data

in packages is sent to the receiver end via. HC-12 transmitter receiver.

d. HC-12:

HC-12 is a wireless module enabling serial port communication which promotes wireless data transfer with an operating frequency range is from 433.4-470 MHz. It is a transceiver module which transmits with sensitivity of 20dBm with the baud rate of 5000bps in the air, and the communication gap is 1km in free space. HC-12 is built with STM8S003F3 microcontroller and the Si4463 transceiver. The Si4463 provides the wireless communication in this circuit. It is embedded with two 64-byte Rx and Tx. STM8S003F3 is a 8-bit microcontroller with 8kB of flash memory and 128 bytes of EEPROM. and a 10-bit ADC. It supports UART, SPI AND I²C communication.

2. Receiving end:

a. Arduino DUE:

Arduino DUE is a 32 bit Atmel SAM3X8E ARM-CORTEX - M3 CPU based microcontroller. It works with a frequency of 84 MHz. possessing 54 digital input/output ports categorizing 12 ports to be used as PWM outputs and remaining 12 to be used as analog inputs. It has a SRAM of 96kB and flash memory of 512kB. It consists of two serial ports which are native serial USB port and programming serial. The native USB serial port supports CDC serial communication using the Serial USB object and is connected directly to the SAM3X8E. As this microcontroller has more number of digital output ports and functions faster, it is preferred at the receiver end.

b. H-Bridge (L298N):

H-Bridge is a driver circuit which controls the DC motors of the robot based on the pulses given by the microcontroller (Arduino DUE). It consists of 4 switches. Activation of diagonal switches leads to flow of current and the direction of current can be reversed by activating the other two switches and switching OFF the other two switches. Change of direction of current through the DC motor will reverse its direction of rotation.

c. Servo Motor (for arm movement):

A servo motor acts linearly as well as in rotational direction to handle an object with great precision. If the system has to be moved to some specified angles or distance, then servo motor can be used. It runs through servo mechanism. On a given command, it only rotates as much required and then wait for next signal to be taken for further action.

III. HARDWARE AND IMPLEMENTATION

a. Movements assigned to the arm:

Below, there are 2 accelerometers and 5 servo motors connected to the supply pins through the Arduino DUE. The accelerometers and servo motors are given a supply of 3.3V and 9V respectively. Following are the degrees of freedom (DOF), no. of joints and no. of links required and the arm diagram:

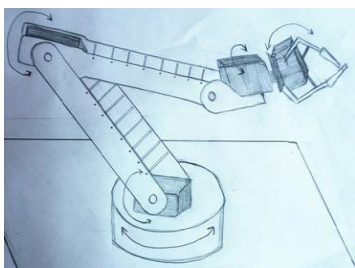


Fig 2 : Sketch representing the robotic arm

The forward and side wise bends assigned to the accelerometer controls all the 4 parts of the arm. The left hand controls the base and the middle link i.e. the shoulder whereas the right hand controls the elbow and the wrist part. Following are the respective directions as given:

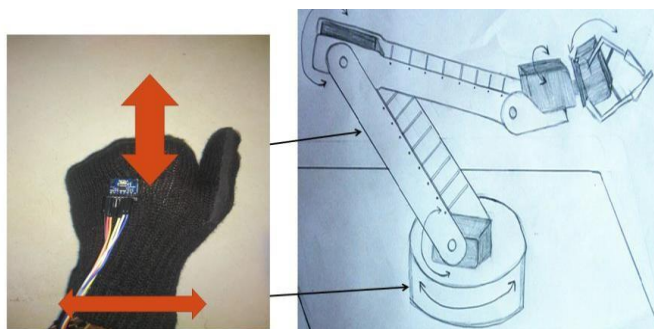


Fig 3. Left hand movements

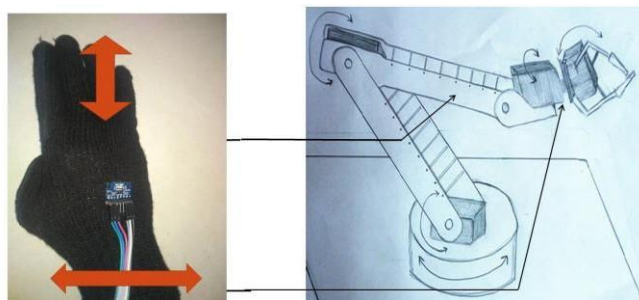


Fig 4. Right hand movements

The gripper is controlled using the flex sensors attached to the glove fingers. The amount of bent of the finger is directly proportional to the amount of bent in the arm. Also, a particular resistance gap is mentioned in the code for the robot to be turned on or off or for the movements to be crisp and clear without any kind of disturbances or mistakes.

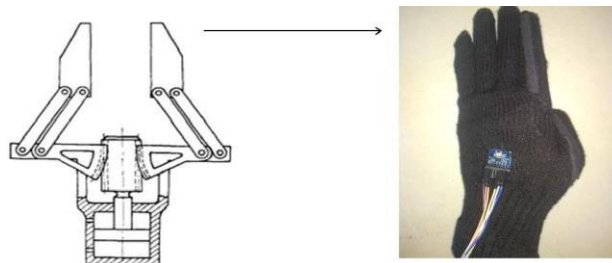


Fig5. Flex sensors controlling the gripper

b. EMP (electromagnetic pulse) gun:

EMP gun is present beneath the robot base which resets the timers which are present in the explosives when arm cannot cut the wires of the bomb. This EMP gun consists of boost converter which boosts up the given voltage to the very higher. The circuit of EMP gun is shown in Fig6.

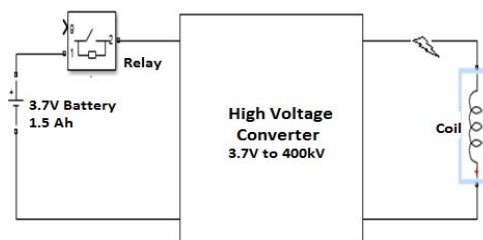


Fig 6. Circuit Diagram of EMP Gun

The Electro-Magnetic Pulses emitted by this gun resets the timers of IED's[9]. This radiations emitted by the robot may also erase the data of the controller of robot, when they get reflected from the IED's. So proper shielding is required in order to avoid this kind of condition.

In addition, that, a camera is equipped on robot which displays the path to the user. This Wi-Fi Camera transmits the path view to mobile, when this camera and mobile are connected to same LAN.

c. Transmitter and receiver:

Flex sensors and accelerometers are connected to the arduino as it collects the outputs of those sensors. Then it gives the input to the HC-12. Accelerometers are connected to arduino using i2c communication. At receiver side HC-12 is connected to Arduino Due so that, it collects the information from HC-12 and operates DC motors, Robotic Arm and EMP gun with the help of relay. Therefore, the overall interfacing of the components and the working process is sketched sequentially in the best cost effective and efficient way. This prototype is implemented to combat the issues aroused in the army field and help the officials to make their surroundings and themselves safe.

IV. RESULT AND DISCUSSION

a. Gesture controlled system:

Gesture recognition is the main base of this system. The figure shown below is the pair of gloves which is equipped with accelerometers and flex sensors. When a specific gesture is made, change in values of flex sensor and accelerometer commands the robot to perform the action.



Fig.7 Glove equipped with sensors

Index and Thumb fingers of both the hands consists of flex sensors and accelerometers are present on the hands such that movement of the hands can be easily recognized.

b. Robot base control via movements:

Base Control of the Robot is the main task of this model. When flex sensor attached to the index finger of right hand is bent, then it activates the robot. This sensor acts a switch of the movement of the robot. When hand is moved forward, backward and side-wards, then the robot moves according to the movement of the right hand. When the hand is bent forward, then the robot moves forward. Similarly if bent backwards, then robot moves backwards. In the same way if the hand is bent side wards then the robot will take turns right or left according to the gesture made.



Fig 8. Robot base movement

c. Robotic Arm control via movements:

Robotic arm movement can be controlled with the help of both the hands. Right hand controls the elbow and wrist part of the arm and left hand controls the base and shoulder part of the arm.

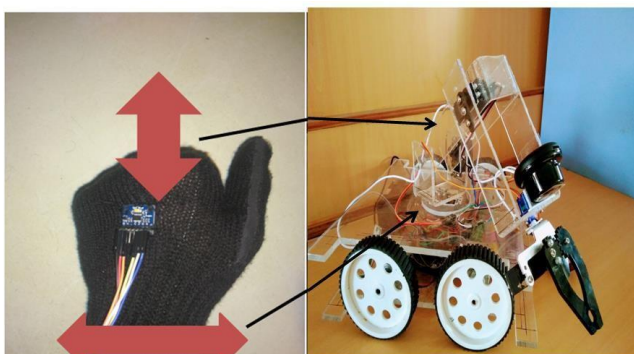


Fig 9. Gesture for the base and shoulder

Change in resistance of flex sensor attached to the left hand will activate the base and shoulder part of the arm. When the hand is moved forward and backward, then shoulder moves front and back where as the side ward movement of hand rotates the base part of the arm. In the same way, change in resistance of Flex sensor attached to the right hand thumb activates the elbow and wrist. Up-Down movement of right hand moves the elbow part and Right-Left movement of that hand rotates the wrist as shown in figure below.

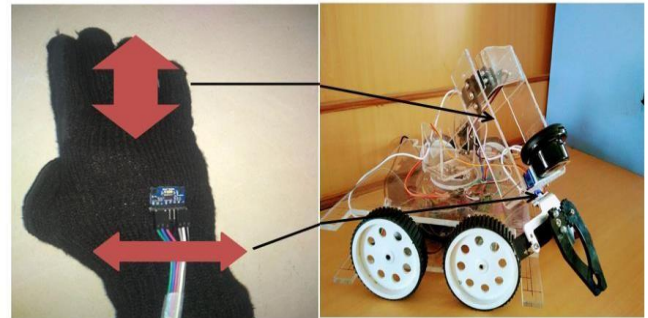


Fig 10. Gesture for the elbow and wrist

c. RF control:

1) Transmitter end:

HC-12 transceiver is used as transmitter and it is interfaced to the arduino with the wires at the user. The figure below gives the idea that how the user will look like if that glove is worn.



Fig11. Glove prototype



Fig12. User with the glove

2) Receiver end:

As explained, the HC-12 present at the receiver side receives the data from the transmitter and gives to controller to perform various actions. The figure 13 explains how the robot appears.

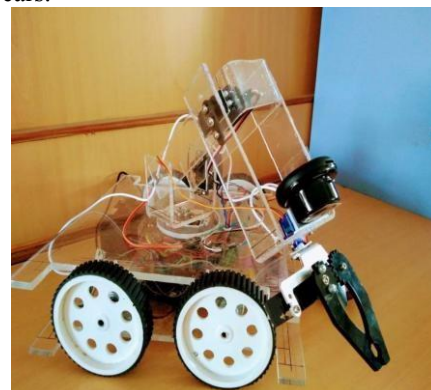


Fig 13. System Prototype

d. EMP gun:

Electromagnetic Gun consists of a battery, relay, a high voltage module which uses the principle of tesla coil and a copper coil. When the flex sensor attached to the left hand thumb is bent, then the controller activates the relay which activates the EMP gun. This resets the counters present in the bomb.



Fig 14. Gesture activating the EMP gun

e. Wi-Fi camera:

This Wi-Fi camera is mounted on the robotic arm of the robot so that direction and gripper control can be done simultaneously as shown below.



Fig15. View from the camera

V. CONCLUSION

Our system tries to overcome the problems and limitations. The most important aspect of our gesture control system is that, it doesn't use complex hardware systems. No complex segmentations and reconstructions are made here. The accelerometer is the main component of the system in robot control system. This project mainly consists of a Hand Gesture Recognition section which controls the robot. The hand gesture recognition section consists of a flex sensor and accelerometer, a microcontroller and a transmitter.

One of the up-gradations of the system was the development of mobile robot systems that were commanded through arm positions but failed to leave an impression since the robotic system couldn't recognize gestures defined through specific temporal patterns. Our system also consists of a Wi-Fi camera which directs the robot through the user and helps controlling the bomb to be diffused virtually. There also another upgradation of EMP gun which acts as a jammer deflecting the nearby electronic devices.

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